Faithful Vision-Language Interpretation via Concept Bottleneck Models

Songning Lai*, Lijie Hu*, Junxiao Wang, Laure Berti-Equille, and Di Wang

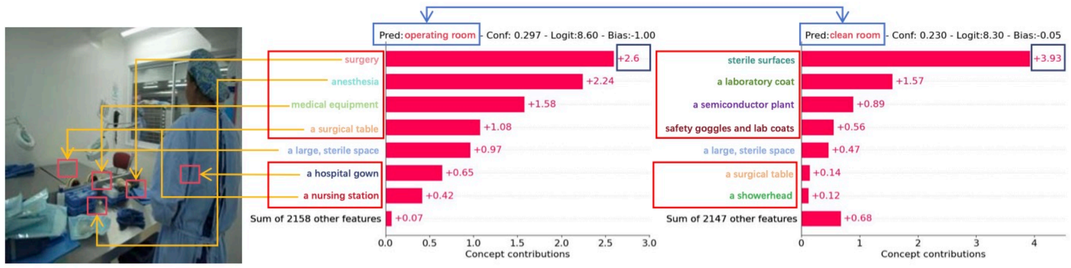
ICLR PRADA Lab

King Abdullah University of Science and Technology

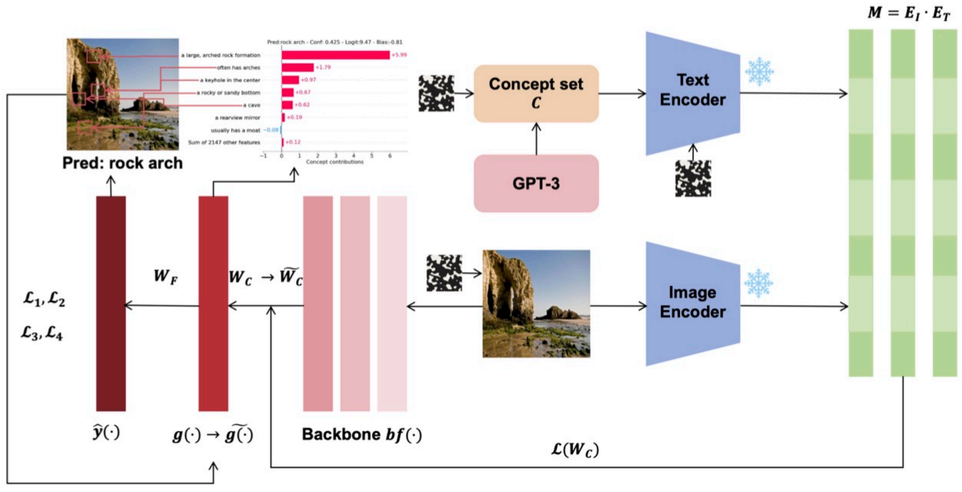
Institut de Recherche pour le Développement FRANCE

Introduction

Traditional Concept Bottleneck Models (CBMs) require substantial manual annotation, which label-free CBM effectively addresses by leveraging factual information from pre-trained models. However, this convenience comes with inherent instability in pre-trained models. We addressed it in this paper.



Definition and Framework



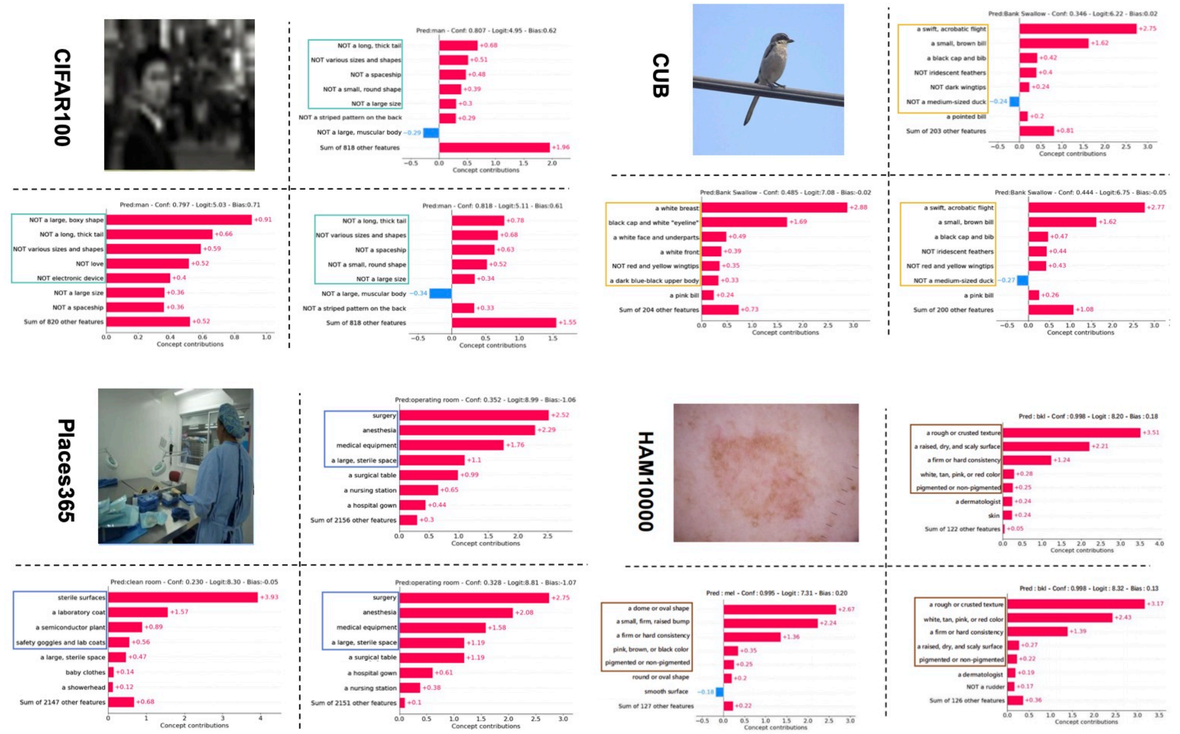
$$\min_{\tilde{W}_c} \mathbb{E}_x[\lambda_1 \underbrace{D(y(x,\tilde{\boldsymbol{c}}), y(x,\boldsymbol{c}))}_{\mathcal{L}_1} + \lambda_2 \underbrace{\mathcal{L}_{k_1}(\tilde{\boldsymbol{g}}(x), \boldsymbol{g}(x))}_{\mathcal{L}_2} + \lambda_3 \underbrace{\max_{||\delta|| \le R_2} D(y(x,\tilde{\boldsymbol{c}}), y(x,\tilde{\boldsymbol{c}}+\boldsymbol{\delta}))}_{\mathcal{L}_3} + \lambda_4 \underbrace{\max_{||\rho|| \le R_1} \mathcal{L}_{k_2}(\tilde{\boldsymbol{g}}(x), \tilde{\boldsymbol{g}}(x)+\boldsymbol{\rho})}_{\mathcal{L}_4}].$$

Definition 1 (Faithful Vision-Language Concept). *Under the same concept space, i.e., under the set of concepts generated by GPT3 at one time, we call a matrix $\tilde{W}_c$ is a $(D, R, \alpha, \beta, k_1, k_2)$-Faithful Vision-Language Concept (FVLC) model for the vanilla concept if it satisfies for any input x:*

- *(Similarity of Explanation) $V_{k_1}(\tilde{\boldsymbol{g}}(x), \boldsymbol{g}(x)) \ge \beta_1$ for some $1 \ge \beta_1 \ge 0$;*
- *(Stability of Explanation) $V_{k_2}(\tilde{\boldsymbol{g}}(x), \tilde{\boldsymbol{g}}(x)+\boldsymbol{\rho}) \ge \beta_2$ for some $1 \ge \beta_2 \ge 0$ and all $\|\boldsymbol{\rho}\| \le R_1$, where $\|\cdot\|$ is a norm and $R_1 \ge 0$;*
- *(Closeness of Prediction) $D(y(x,\tilde{\boldsymbol{c}}), y(x,\boldsymbol{c})) \le \alpha_1$ for some $\alpha_1 \ge 0$, where D is some probability distance or divergence;*
- *(Stability of Prediction) $D(y(x,\tilde{\boldsymbol{c}}), y(x,\tilde{\boldsymbol{c}}+\boldsymbol{\delta})) \le \alpha_2$ for all $\|\boldsymbol{\delta}\| \le R_2$, where D is some probability distance or divergence, $\|\cdot\|$ is a norm and $R_2 \ge 0$,*

where $\tilde{\boldsymbol{g}}(x) = \tilde{W}_c bf(x)$, $y(x,\boldsymbol{c}) = W_F g(x)$, and $y(x,\tilde{\boldsymbol{c}}) = W_F \tilde{\boldsymbol{g}}(x)$, $y(x,\tilde{\boldsymbol{c}}+\boldsymbol{\delta}) = W_F(\tilde{\boldsymbol{g}}(x)+\boldsymbol{\delta})$. For any given x, $\tilde{c} = \tilde{\boldsymbol{g}}(x)$ is a $(D, R, \alpha, \beta, k_1, k_2)$-FVLC. Here, $\alpha = \min\{\alpha_1, \alpha_2\}$, $\beta = \max\{\beta_1, \beta_2\}$, and $R = \min\{R_1, R_2\}$.

Evaluating Interpretability and Utility



Stability Evaluation

Method	CIFAR10	CIFAR100	CUB	Places365
Standard (No interpretability)	88.80%	70.10%	76.70%	48.56%
P-CBM (CLIP)	84.50%	56.00%	N/A	N/A
Label-free CBM	86.32%	65.42%	74.23%	43.63%
WP1(5%) - base	86.47%	65.13%	74.08%	43.57%
WP1(5%) - **FVLC**	86.34%	**65.43%**	73.96%	43.67%
WP1(10%) - base	86.25%	65.09%	73.97%	43.67%
WP1(10%) - **FVLC**	86.39%	64.90%	73.92%	43.62%
WP2 - base	86.41%	65.16%	73.96%	43.54%
WP2 - **FVLC**	86.22%	65.34%	**74.44%**	**44.55%**
IP - base	86.62%	65.36%	74.39%	43.64%
IP - **FVLC**	**86.88%**	65.29%	74.01%	43.71%
WP1(5%)+WP2 - base	86.49%	65.17%	73.90%	43.67%
WP1(5%)+WP2 - **FVLC**	86.43%	65.33%	73.92%	43.49%
WP1(10%)+WP2 - base	86.30%	64.87%	73.82%	43.61%
WP1(10%)+WP2 - **FVLC**	86.38%	65.06%	74.01%	43.44%
WP1(10%)+WP2+IP - base	85.96%	64.41%	73.74%	43.32%
WP1(10%)+WP2+IP - **FVLC**	86.70%	65.14%	74.36%	43.46%

Method	CIFAR10		CIFAR100		CUB		Places365	
	TCPC	TOPC	TCPC	TOPC	TCPC	TOPC	TCPC	TOPC
WP1(5%) - base	1.55E-01	6.32E-02	1.01E-01	7.17E-02	1.26E-01	1.85E-01	1.59E-01	6.40E-02
WP1(5%) - **FVLC**	**1.12E-03**	**8.55E-03**	**2.81E-03**	**4.51E-03**	**1.05E-02**	**1.50E-03**	**1.38E-03**	**1.30E-03**
WP1(10%) - base	1.99E-01	8.36E-02	1.94E-01	1.31E-01	2.32E-01	3.41E-01	2.26E-01	1.14E-01
WP1(10%) - **FVLC**	**1.19E-03**	**7.40E-03**	**3.67E-03**	**4.55E-03**	**1.19E-02**	**1.53E-03**	**1.39E-03**	**1.25E-03**
WP2 - base	1.53E-01	4.99E-02	1.36E-01	6.67E-02	1.43E-01	1.73E-01	1.40E-01	6.37E-02
WP2 - **FVLC**	**1.10E-02**	**8.72E-03**	**3.35E-03**	**4.55E-03**	**1.05E-02**	**1.53E-03**	**1.55E-03**	**1.29E-03**
IP - base	1.68E-01	6.28E-02	1.38E-01	8.81E-02	1.71E-01	2.23E-01	1.73E-01	8.09E-02
IP - **FVLC**	**8.02E-03**	**8.29E-03**	**3.24E-03**	**4.56E-03**	**1.04E-02**	**1.59E-03**	**1.50E-03**	**1.25E-03**
WP1(5%)+WP2 - base	1.85E-01	3.50E-02	1.28E-01	6.65E-02	1.44E-01	1.79E-01	1.60E-01	6.32E-02
WP1(5%)+WP2 - **FVLC**	**1.20E-02**	**7.46E-03**	**3.67E-03**	**4.55E-03**	**9.81E-02**	**1.51E-03**	**1.54E-03**	**1.28E-03**
WP1(10%)+WP2 - base	1.17E-01	8.62E-02	1.93E-01	1.32E-01	1.76E-01	3.45E-01	2.52E-01	1.17E-01
WP1(10%)+WP2 - **FVLC**	**1.18E-02**	**9.41E-03**	**2.06E-02**	**1.44E-02**	**1.87E-02**	**3.79E-02**	**2.74E-02**	**1.18E-02**
WP1(10%)+WP2+IP - base	1.36E-01	1.05E-01	2.22E-01	1.55E-01	1.95E-01	3.54E-01	2.62E-01	1.44E-01
WP1(10%)+WP2+IP - **FVLC**	**1.43E-02**	**1.11E-02**	**2.39E-02**	**1.77E-02**	**2.21E-02**	**4.54E-02**	**3.35E-02**	**1.34E-02**

Ablation Study

Method	Setting			CIFAR10		CIFAR100		CUB		Places365	
	$\mathcal{L}_2$	$\mathcal{L}_3$	$\mathcal{L}_4$	TCPC	TOPC	TCPC	TOPC	TCPC	TOPC	TCPC	TOPC
WP1(10%) - FVLC				1.99E-01	8.36E-02	1.94E-01	1.31E-01	2.32E-01	3.41E-01	2.26E-01	1.14E-01
	✓			2.09E-02	3.14E-02	2.81E-02	4.88E-02	4.08E-01	7.56E-02	4.69E-02	6.14E-02
		✓		1.80E-02	1.79E-02	2.01E-02	2.85E-02	3.77E-01	4.50E-02	4.48E-02	3.68E-02
			✓	4.78E-03	3.11E-02	1.67E-02	2.19E-02	5.69E-02	5.52E-03	6.04E-03	4.98E-03
	✓	✓		1.67E-02	1.11E-02	5.39E-02	6.85E-02	1.69E-01	2.12E-02	1.98E-02	1.81E-02
	✓		✓	1.65E-03	1.01E-02	5.08E-03	6.43E-03	1.61E-02	2.14E-03	1.95E-03	1.72E-03
		✓	✓	1.63E-03	1.02E-02	5.04E-03	6.27E-03	1.63E-02	2.16E-03	1.94E-03	1.71E-03
	✓	✓	✓	**1.19E-03**	**7.40E-03**	**3.67E-03**	**4.55E-03**	**1.19E-02**	**1.53E-03**	**1.39E-03**	**1.25E-03**
WP2 - FVLC				1.53E-01	4.99E-02	1.36E-01	6.67E-02	1.43E-01	1.73E-01	1.40E-01	6.37E-02
	✓			7.62E-02	2.29E-02	2.02E-02	2.08E-02	6.09E-02	5.22E-03	1.04E-02	5.31E-03
		✓		5.37E-02	1.84E-02	1.46E-02	1.16E-02	6.80E-02	2.95E-03	1.06E-02	2.36E-03
			✓	5.19E-02	1.46E-02	1.31E-02	9.34E-03	5.30E-02	3.24E-03	6.83E-03	2.12E-03
	✓	✓		4.57E-02	1.82E-02	1.45E-02	1.01E-02	5.01E-02	3.46E-03	6.30E-03	2.94E-03
	✓		✓	2.73E-02	6.96E-03	7.10E-03	4.60E-03	2.42E-02	1.91E-03	3.97E-03	2.36E-03
		✓	✓	2.47E-02	8.00E-03	5.29E-03	5.25E-03	2.40E-02	1.70E-03	3.82E-03	1.96E-03
	✓	✓	✓	**1.10E-02**	**8.72E-03**	**3.35E-03**	**4.55E-03**	**1.05E-02**	**1.53E-03**	**1.55E-03**	**1.29E-03**
IP - FVLC				1.68E-01	6.28E-02	1.38E-01	8.81E-02	1.71E-01	2.23E-01	1.73E-01	8.09E-02
	✓			6.39E-02	2.55E-02	2.56E-02	3.57E-02	8.04E-02	1.27E-02	1.16E-02	9.80E-03
		✓		3.82E-02	3.91E-02	1.53E-02	2.16E-02	4.77E-02	7.66E-03	6.99E-03	5.88E-03
			✓	1.63E-02	1.73E-02	6.35E-03	9.35E-03	2.14E-02	3.30E-03	3.01E-03	2.59E-03
	✓	✓		2.04E-02	2.10E-02	8.35E-03	1.15E-02	2.67E-02	4.04E-03	3.77E-03	3.18E-03
	✓		✓	1.23E-02	1.27E-02	5.09E-03	6.94E-03	1.65E-02	2.45E-03	2.35E-03	1.94E-03
		✓	✓	1.26E-02	1.09E-02	5.07E-03	7.00E-03	1.59E-02	2.53E-03	2.29E-03	1.95E-03
	✓	✓	✓	**8.02E-03**	**8.29E-03**	**3.24E-03**	**4.56E-03**	**1.04E-02**	**1.59E-03**	**1.50E-03**	**1.25E-03**
WP1(10%)+WP2 - FVLC				1.17E-01	8.62E-02	1.93E-01	1.32E-01	1.76E-01	3.45E-01	2.52E-01	1.17E-01
	✓			5.97E-02	6.28E-02	1.43E-01	9.94E-02	1.24E-01	2.58E-01	1.86E-01	8.24E-02
		✓		3.76E-02	6.34E-02	1.48E-01	1.03E-01	1.21E-01	2.63E-01	1.90E-01	8.13E-02
			✓	3.52E-02	2.54E-02	6.03E-02	4.13E-02	5.08E-02	1.07E-01	7.63E-02	3.31E-02
	✓	✓		9.21E-02	5.45E-02	1.28E-01	8.56E-02	1.04E-01	2.24E-01	1.60E-01	6.82E-02
	✓		✓	3.15E-02	2.90E-02	6.88E-02	4.71E-02	5.76E-02	1.21E-01	8.86E-02	3.72E-02
		✓	✓	1.53E-02	1.18E-02	2.78E-02	1.89E-02	2.37E-02	4.88E-02	3.62E-02	1.55E-02
	✓	✓	✓	**1.18E-02**	**9.41E-03**	**2.06E-02**	**1.44E-02**	**1.87E-02**	**3.79E-02**	**2.74E-02**	**1.18E-02**
WP1(10%)+WP2+IP - FVLC				1.36E-01	1.05E-01	2.22E-01	1.55E-01	1.95E-01	3.54E-01	2.62E-01	1.44E-01
	✓			7.37E-02	5.75E-02	1.29E-01	9.24E-02	1.21E-01	2.34E-01	1.74E-01	7.30E-02
		✓		7.85E-02	5.85E-02	1.23E-01	9.17E-02	1.14E-01	2.43E-01	1.70E-01	6.80E-02
			✓	4.59E-02	3.67E-02	7.62E-02	5.57E-02	6.74E-02	1.41E-01	1.08E-01	4.23E-02
	✓	✓		5.81E-02	4.52E-02	9.69E-02	7.11E-02	8.99E-02	1.83E-01	1.34E-01	5.49E-02
	✓		✓	5.84E-02	4.53E-02	9.75E-02	7.17E-02	8.94E-02	1.85E-01	1.36E-01	5.41E-02
		✓	✓	2.91E-02	2.31E-02	4.85E-02	3.56E-02	4.46E-02	9.33E-02	6.78E-02	2.81E-02
	✓	✓	✓	**1.43E-02**	**1.11E-02**	**2.39E-02**	**1.77E-02**	**2.21E-02**	**4.54E-02**	**3.35E-02**	**1.34E-02**

Contact

Corresponding to lijie.hu@kaust.edu.sa